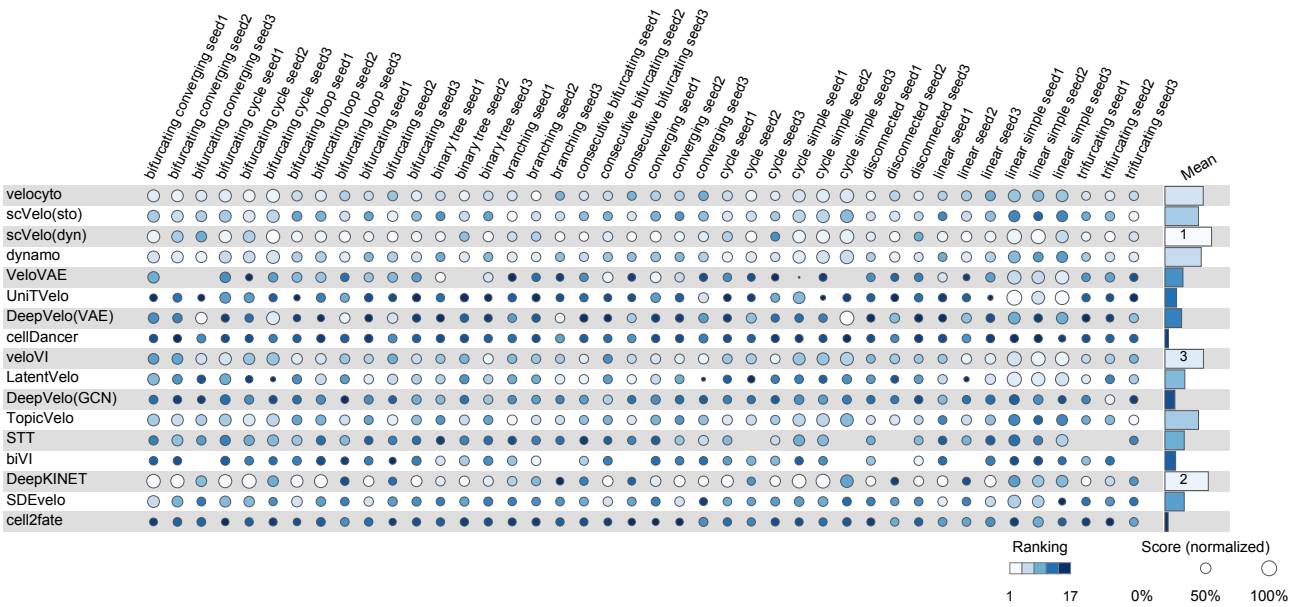
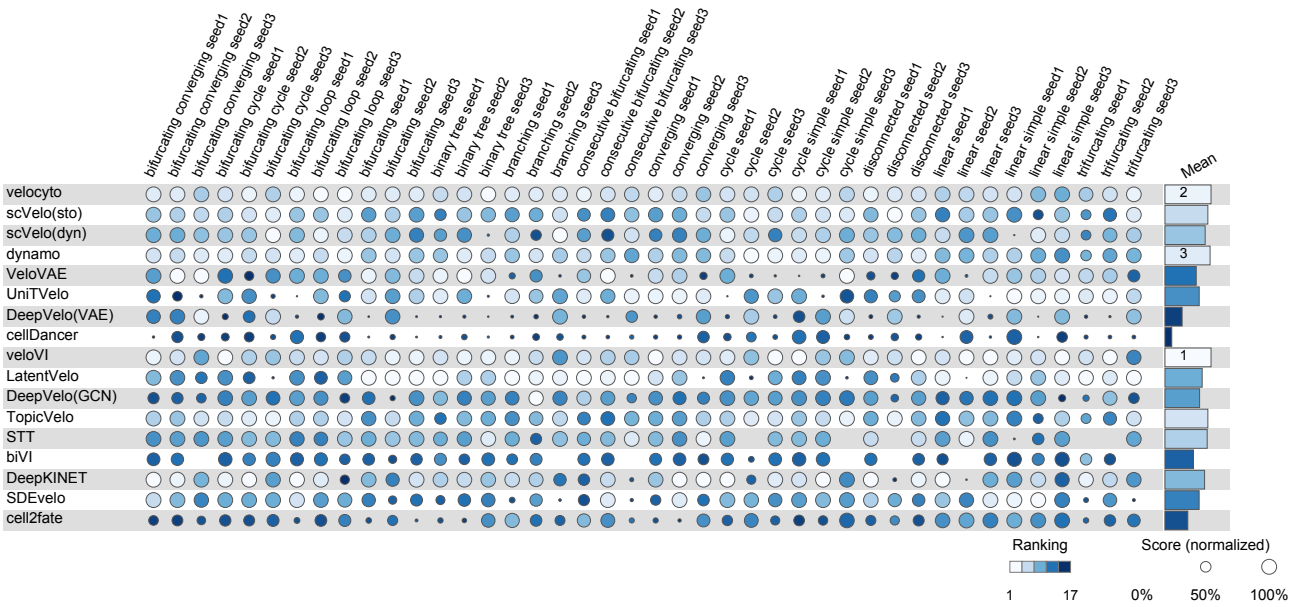
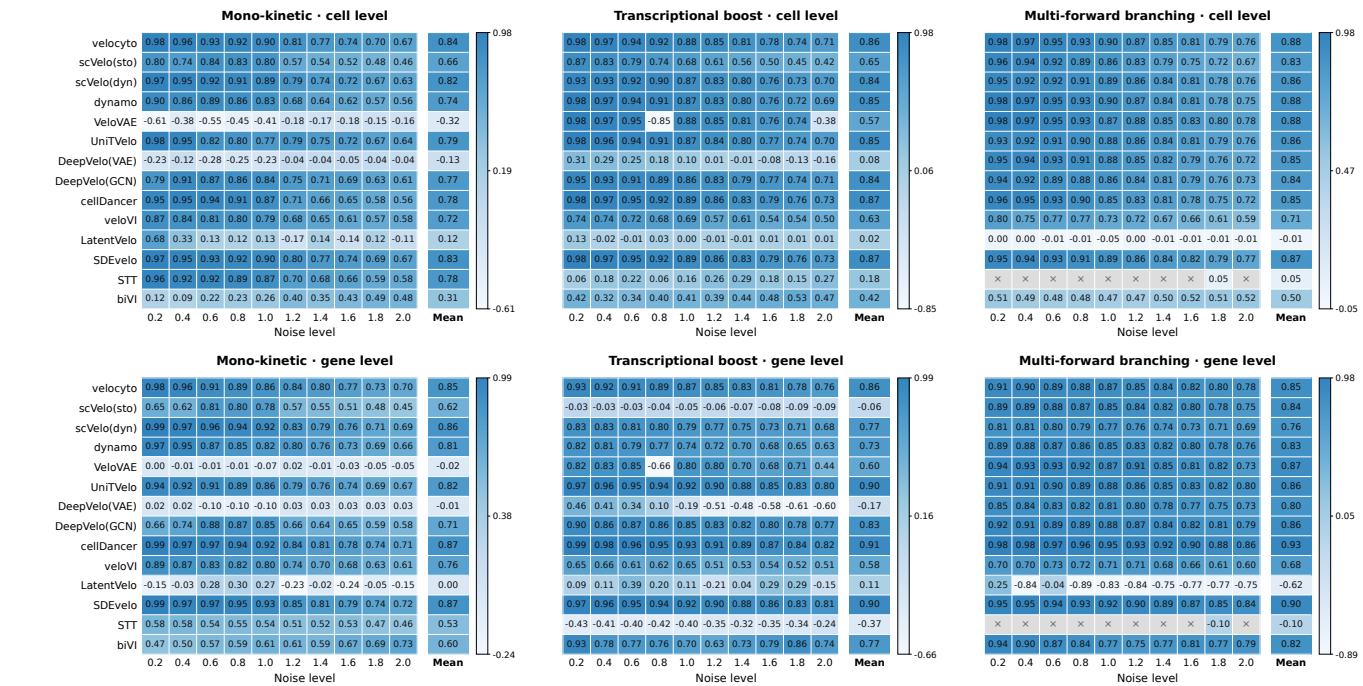
a


Cosine Similarity of High-dim RNA Velocity in Gene Level on Dyngen Simulation Datasets (n=42)
bifurcating converging seed1
bifurcating converging seed2
bifurcating converging seed3
bifurcating cycle seed1
bifurcating cycle seed2
bifurcating cycle seed3
bifurcating loop seed1
bifurcating loop seed2
bifurcating loop seed3
bifurcating seed1
bifurcating seed2
bifurcating seed3
binary tree seed1
binary tree seed2
binary tree seed3
branching seed1
branching seed2
branching seed3
consecutive bifurcating seed1
consecutive bifurcating seed2
consecutive bifurcating seed3
converging seed1
converging seed2
converging seed3
cycle seed1
cycle seed2
cycle seed3
cycle simple seed1
cycle simple seed2
cycle simple seed3
disconnected seed1
disconnected seed2
disconnected seed3
linear seed1
linear seed2
linear seed3
linear simple seed1
linear simple seed2
linear simple seed3
trifurcating seed1
trifurcating seed2
trifurcating seed3
Mean
velocyto
scVelo(sto)
scVelo(dyn)
dynamo
VeloVAE
UniTVelo
DeepVelo(VAE)
cellDancer
veloVI
LatentVelo
DeepVelo(GCN)
TopicVelo
STT
biVI
DeepKINET
SDEvelo
cell2fate
1
3
2
Ranking
Score (normalized)
1
17
0%
50%
100%

b


Cosine Similarity of Low-dim RNA Velocity in Cell Level on Dyngen Simulation Datasets (n=42)
bifurcating converging seed1
bifurcating converging seed2
bifurcating converging seed3
bifurcating cycle seed1
bifurcating cycle seed2
bifurcating cycle seed3
bifurcating loop seed1
bifurcating loop seed2
bifurcating loop seed3
bifurcating seed1
bifurcating seed2
bifurcating seed3
binary tree seed1
binary tree seed2
binary tree seed3
branching seed1
branching seed2
branching seed3
consecutive bifurcating seed1
consecutive bifurcating seed2
consecutive bifurcating seed3
converging seed1
converging seed2
converging seed3
cycle seed1
cycle seed2
cycle seed3
cycle simple seed1
cycle simple seed2
cycle simple seed3
disconnected seed1
disconnected seed2
disconnected seed3
linear seed1
linear seed2
linear seed3
linear simple seed1
linear simple seed2
linear simple seed3
trifurcating seed1
trifurcating seed2
trifurcating seed3
Mean
velocyto
scVelo(sto)
scVelo(dyn)
dynamo
VeloVAE
UniTVelo
DeepVelo(VAE)
cellDancer
veloVI
LatentVelo
DeepVelo(GCN)
TopicVelo
STT
biVI
DeepKINET
SDEvelo
cell2fate
2
3
1
Ranking
Score (normalized)
1
17
0%
50%
100%

c


Mono-kinetic · cell level
Transcriptional boost · cell level
Multi-forward branching · cell level
Mono-kinetic · gene level
Transcriptional boost · gene level
Multi-forward branching · gene level
velocyto
scVelo(sto)
scVelo(dyn)
dynamo
VeloVAE
UniTVelo
DeepVelo(VAE)
DeepVelo(GCN)
cellDancer
veloVI
LatentVelo
SDEvelo
STT
biVI
0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0
Mean
Noise level

Extended Data Fig. 4 | Cosine similarity on simulation datasets. a, Gene-level cosine similarity between predicted high-dimensional RNA velocity and simulated truth on 42 dyngen datasets. **b,** Cell-level cosine similarity in PCA2 on the same 42 datasets. Alternating grey rows separate methods in the blue bubble matrices, with Median/Mean bars. Dot area is normalized within each panel using the global minimum minus 0.01 and maximum; blue fill shows within-dataset rank (lighter is better). Median/Mean summarize raw dataset scores; their bar widths share a minimum-minus-0.01/max display scale within each panel. Numbers mark the top six ranks, with ties retained. Display normalization does not change the raw cosine values. **c,** Full cellDancer noise series (0.2–2.0 in steps of 0.2), displayed as heatmaps for three scenarios at cell and gene levels. Within each heatmap, rows are methods and columns are noise levels. Each entry is the median cosine over valid cells or genes in one simulated dataset, using the common finite genes shared by available non-exploratory methods within that dataset. Each heatmap uses its own linear color scale from its lowest to highest available median cosine, including the Mean column, with a separate color bar. Darker blue indicates higher agreement within that heatmap; identical colors across heatmaps do not imply identical scores. Numeric annotations retain the original cosine values. The rightmost Mean column is the equally weighted arithmetic mean of available dataset medians across noise levels for that method, scenario and evaluation level. Each number is rounded to two decimals. All means cover ten noise conditions except STT in forward branching, which covers only one (noise 1.8). Grey crosses denote missing results; missing values are not imputed, and zero-norm directions are excluded. biVI is exploratory on continuous U/S input and uses the same gene set. Shared gene counts are 187–1,000 in mono-kinetic conditions and 1,000 otherwise. Each scenario/noise condition has one dataset; cross-noise means are descriptive summaries, not independent replicate estimates. Varying gene sets and saved preprocessing limit attribution solely to noise.